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# Give me your undivided Attention! - Balancing Persuasion and Ethics in Digital Design

Ethics of Artificial Intelligence by  
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# 1 Introduction

In the age of civilization rooted in technological advancements, everyday life is strongly influenced by a constantly expanding knowledge base and thus by information-rich systems. Information takes on enormous value. Every click, scroll and share forms the breeding ground of data. Designers and developers of these systems are now faced with the task of creating products that, on the one hand, capture the attention of users in a never-ending stream of information and, on the other, ensure that these efforts do not overstep the boundaries of ethical principles.

It all revolves around the central question of how digital products can be designed to promote positive behavioral change without drifting into manipulative practices that undermine users' freedom of choice. How to create ethical design that supports and promotes autonomy while engaging, informing and supporting the user in an information-rich digital world.

At a time when attention is considered one of the most limited resources, this work explores the mechanisms of the attention economy and how it influences the development of technologies that aim to maximize our time and interest. At the same time, it sheds light on the ethical considerations that need to be factored into the design process in order to strike a balance between effectiveness and ethical responsibility. From the subtle art of "nudging" to the explicit techniques of persuasion, from the principles of value-sensitive design to the challenges of AI-powered persuasion, this work aims to create a deep understanding of how digital design shapes our perceptions, our choices, and ultimately our behavior, and how this power must be used with great ethical care.

"Give me your undivided Attention!" is a call to broaden the discussion about digital design - from a focus on technical and economic aspects to a comprehensive consideration of the ethical dimensions inherent in the design of digital experiences and guidelines for a human-driven design approach in the digital world.

## 2 Foundation

### 2.1 Attention Economy

The idea of the concept of the attention economy was initially explored by H. Simon 1971 [1]. For him "in an information-rich world, the wealth of information means a dearth of something else: a scarcity of whatever it is that information consumes. What information consumes is rather obvious: it consumes the attention of its recipients" [1]. In an information-rich economy the resource of attention is highly valuable due to its inherent limitations. Furthermore attention is also measurable, allowing precise calculation of activities which leads to enabling monetization, thereby attributing economic value to attention [2]. The success of economic activity is therefore based on catching the attention of the perceivers of the information. The development of information and communication technologies revealing that the true commodity being offered and sold is the public's attention [3]. However, the business model of the attention economy also takes into account the loss of attention. Silicon Valley companies, especially social media companies, have focused their efforts on knowing even the smallest movement of users in order to develop services and apps designed to get users to spend more time with them [4]. It is primarily the data that interests them, which are then used not just for service improvement but especially for monetary purposes such as advertising revenue [5]. Aza Raskin even compares the trade in human attention and thus the trade in human behavior to the human slave trade, arguing that "the business model is not just selling our attention, but selling the change in our behaviors, beliefs, and biases" [6]. As is widely known "if you are not paying for the product, you are the product" which, in the context of the attention economy, refers to the attention provided for free.

### 2.2 Persuasive Design Techniques

The attention economy works on the premise that progressions in technology, appealing interface designs, and complex notification frameworks empower the control of human attention [7]. Persuasive technologies are a technique for controlling human attention and thus human behavior in the digital world. Oinas-Kukkonen and Harjumaa [8] define a persuasive system as "a computerized software or Information system designed to reinforce, change or shape attitudes or behaviors or both without using coercion or deception." According to Fogg's theory about persuasion [9]. There are three different types of persuasion: interpersonal persuasion, computer-mediated persuasion and human-computer persuasion, where the latter is meant when interacting with technology. It is not always explicitly stated who the persuader is, as it is assumed that computers have no intentions of their own [10]. Products in the attention economy use persuasive techniques in their design that are similar to the techniques used in the non-digital world.

| Technique    | Description                                                                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tailoring    | Information is tailored to the user’s potential needs, interests, personality or other factors, which increases the user’s motivation to engage with the content [11].                                                                       |
| Tunneling    | Users are guided through a process or experience by the presentation and visualization of appropriate information [11]. They engage once they have entered the process, often voluntarily, as the entry barrier is very low [12].            |
| Reduction    | Complex activities are reduced to a few simple steps, in some cases a single step. Non-relevant, unnecessary steps are removed. The simpler the more successful the target behavior, as users often prefer the path of least resistance [13] |
| Surveillance | The use of technology to monitor the behavior of other people in order to change their behavior in a certain way. Knowing that they are being watched makes users behave differently [12].                                                   |

Table 1: Sample of the most common Persuasive Design techniques, first introduced by Fogg [9].

In addition to the approach of Persuasive Design, there is also Digital Nudging as an specific approach within this field of influencing user behavior. Digital nudging refers to an approach that uses user interface elements to influence user decision making and that shapes the behavior of individuals in digital decision environments. The focus is on enhancing decision making by reducing or removing barriers to cause a desired behavior [14].

| Technique  | Description                                                                                                                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Incentives | In order to make planned incentives effective for the user, the design can increase their conspicuousness to draw the individual’s attention to them [15].                                                                                        |
| Feedback   | The best way to help people improve their performance is to give them feedback. Giving users feedback when they perform well or make mistakes raises their awareness and encourages behavioral change [14].                                       |
| Mapping    | Mapping refers to a design principle that aims to support and enhance an individual’s ability to identify and select options. One concept is to make information more understandable [15].                                                        |
| Defaults   | All choices presentations contain a default setting in their design, which according to Johnson et al. [16] is chosen more frequently than the other options. This is again due to the psychological principle of ”the path of least resistance”. |

Table 2: Sample of the most common Digital Nudging techniques.

Both persuasive design and digital nudging have the same basic goal of influencing users’ behavior, attitudes and decisions towards a desired goal. However, persuasive design focuses on attitude-oriented design. Digital nudging, on the other hand, focuses on

decision-oriented design. Nudging can facilitate behavior change by removing obstacles rather than trying to push or force people in a certain direction [17]. Consequently, the theory of nudging goes hand in hand with the persuasive design principle of reduction. The overlap between the two strategies is expressed as free choices, change-oriented, and path of least resistance [11].

## 2.3 AI-driven persuasion

With the advancement of technologies and their ubiquitous involvement and value extraction in human communication, Artificial Intelligence (AI) is increasingly taking on a new form as a persuader, influencing the perceptions, attitudes and behavior of users in different areas of human life. AI-driven or AI-based persuasion is defined as “a symbolic process in which a communicative-AI entity generates, augments, or modifies a message—designed to convince people to shape, reinforce, or change their responses—that is transmitted to human receivers” [18]. Due to the technological nature of AI, AI as a persuader has its own characteristics. Firstly, it takes on a dual role in the persuasive communication process, both as a sender and as a relationship partner, which results in a relational interaction. In addition, she mimics human cognitive processes. Its persuasive actions are based on the ability to analyze collected information, which in comparison exceeds the human capacity for data processing by far [19]. Persuasion varies with AI complexity. “Thin” AI, with limited cues and indirect communication, struggles in persuasion due to minimal engagement. Conversely, “Thick” AI, rich in cues, offering direct and interactive communication, is more effective in persuasion by actively engaging users [18]. Well-known paradigms to analyze AI-based persuasion are, for example, the CASA (Computers Are Social Actors) paradigm, which considers how people perceive computers as social actors and interact with them by attributing human-like characteristics to them. Or the MAIN (Modality, Agency, Interactivity and Navigability) paradigm, which focuses on the modality, agency, interactivity and navigability of AI-based persuasion and examines how differences in AI complicate the persuasion process [18].

## 3 Ethical Considerations on Persuasive Design

### 3.1 Manipulation VS Persuasion

It is important to distinguish between persuasion, manipulation and everything in between in order to understand the dilemma between influencing behavior in a way that respects autonomy and manipulating decisions covertly. *Persuasion* is a direct appeal to the decision-making capacity of an individual, who is still free to decide after having had the opportunity to understand and validate the information presented to themselves [20]. From an ethical (and also legal) point of view, this is justifiable as the individual can weigh the information against their personal beliefs, desires and values without unknown or unwanted outside influences. And all this before making a decision regarding a possible change in behavior. If the available options of a decision-making process are restricted due to the user's weaknesses and vulnerabilities, this restriction of autonomy is called *coercion*. Although the user has a range of choices, as with persuasion, and knowing the good intention of the persuasion techniques, coercion is nevertheless unethical as it prevents the individual from considering all the information and allowing themselves to make a free, well-informed and thoughtful decision [20]. Coercion is an active attempt to control the outcome of the decision-making process through irresistible incentives, which means that no conscious decision can be made. People are treated as mere resources without respecting their legal autonomy and human rights [21]. If false information or half-truths are presented to the uninformed user as the basis for a decision, this is called *deception*. People's vulnerability is exploited in such a way that they cannot influence the decision-making process and subsequent behavior autonomously in any way. Strictly controlled deception is controversial but permissible. However, deceptive practices are neither ethically nor legally permissible [22]. The nature of *manipulation*, on the other hand, lies in the aim of exerting unnoticed influence in order to induce a desired behavior that is not necessarily in the best interests of the user [9]. It not only undermines the individual's ability to make decisions, but also exploits human vulnerabilities in a similar way to coercion, which, unlike manipulation, is used directly rather than subtly. By undermining individual autonomy through covert manipulative techniques, manipulation practices have the ability to slyly alter individual and group decisions and behaviors. "Practices in digital interfaces that steer, deceive, coerce, or manipulate consumers into making choices that often are not in their best interests" [23] are known as *Dark Patterns*. Persuasive technologies blur the lines between persuasion and manipulation to an unprecedented degree [24], thereby attacking the autonomy, dignity and freedom of choice of the individual and so the voluntariness of change.

### 3.2 Beneficence VS Exploitation

Persuasive Technologies are also always confronted with the balancing act between acting in the best interest of the user and exploiting the user for the benefit of the system or its creators, which puts the well-being of the user second. By definition, the creators of such systems have a strong control over the outcome of the system through their intention. Target behavior is primarily an intended positive behavioral outcome. Surprising behavior is a positive behavioral outcome that was not intended but provides an additional benefit and contributes to the well-being of the user. Backfiring contains multiple negative outcomes (side effects) that are triggered regardless of the design [25]. Studies show that users can be persuaded with persuasive technologies to make decisions that are in their own benefit [26]. Easy access to information is essential, as is the transparency of the decision-making process through which they are guided. Then applications of persuasive technologies can make a good contribution to human development, e.g. in health, education, sustainability, but also error prevention. Creators who act with the intention of hiding their true intentions for the users take the well-being and interests of the user into account only insufficiently or not at all [27]. So-called Dark patterns emerge [28].

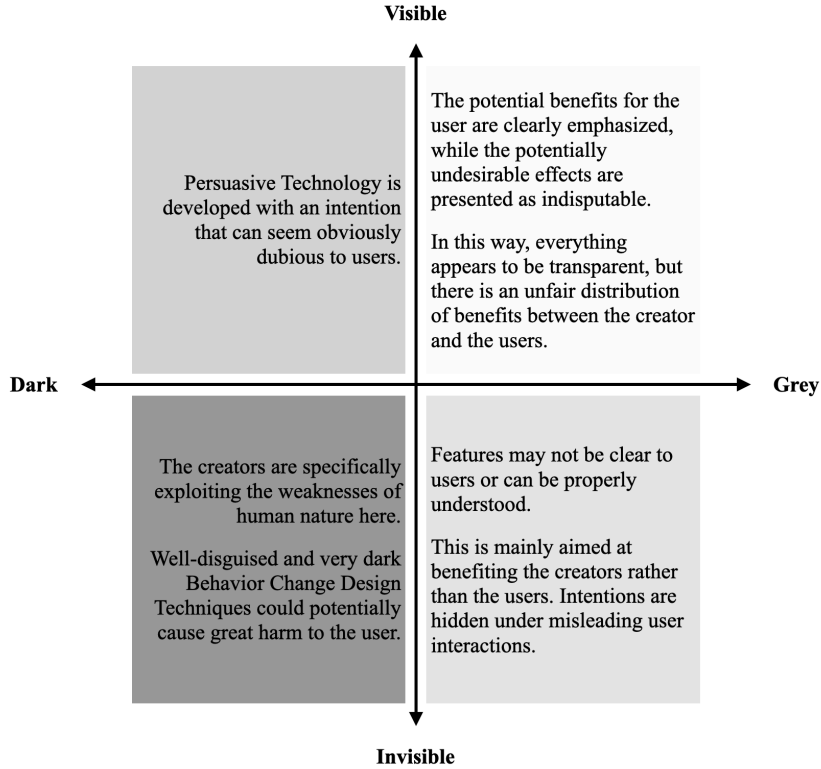


Figure 1: Classification of dark patterns proposed by Nyström and Stibe [28], which represent the use of persuasive design techniques that are intentionally designed by the creator of the system with a negative outcome/intention.

In an attention economy, it is important to elicit behavioral information from users that can be resold and thus benefit the creator. Dark patterns can vary in intensity, they



can have varying degrees of impact on users and their desired outcomes. Does persuasive design reinforce an addiction or is it "just" collecting user data [28]. Honesty and transparency are rarely disclosed to the user, regardless of the intention. Social media platforms in particular use persuasive technologies and deny this dilemma due to their business model.

### 3.3 AI-driven VS Human-driven

While human-driven behavior change still consists of the system being created by humans for humans, AI-driven persuasion tools consist of AI algorithms that control and maintain the persuasion system. The resulting actions to influence content are based on the analysis of the information collected [29]. In contrast to the human-driven approach, access to persuasion is now shifting to those who can use AI-supported technology most effectively. Not only is the control of the information flow no longer in human hands, but it can no longer be fully understood due to the limitations of human cognitive abilities. Person-centered persuasion is now possible on a large scale through big data, which without human judgment can lead to an exacerbated misperception of the popularity of ideas [30].

| <b>Human-driven</b>                              | <b>AI-driven</b>                                             |
|--------------------------------------------------|--------------------------------------------------------------|
| Physical appearance strengthens trust            | Generation and selection of most effective content           |
| Modality in authenticity with contextual nuances | No reputational concerns, ability to take on different roles |
| Ability to change conformity                     | Unlimited social persistence in the interaction              |

Table 3: Strengths of the individual techniques [19][30].

A study by Huang and Wang [19] showed that AI agents were just as persuasive as humans in terms of overall persuasion performance. However, they were less effective than humans in forming specific behavioral intentions, which can be explained by skepticism towards algorithmic decision-making [31]. However, not to the extent that the user reacts differently to AI-driven and human content. The issue of moral justifiability is more about the act of persuasion than the AI that performs it. Which is directly related to the difficulty of identifying clear responsibilities for the persuader. "Since the persuader is not "liable for what he/she/it says, the persuader retreats in the background and the persuasion act remains in the foreground" [32]. The user therefore does not care whether an AI system or a human is lying in order to persuade them. If AI systems become more persuasive than almost all of humanity, this could become a threat to human autonomy in general and, more specifically, lead to a loss of human control. The uncontrolled promotion of Ai-driven techniques leads to a serious degradation of discourse between

humans through e.g. degradation of truthfulness and subsequent loss of trust. Unwanted or false beliefs can spread even faster as systems depend on the beliefs of very small groups. The surveillance and intrusion of society would throw sovereignty into a serious crisis [30].

## 4 Perspectives on Ethical Design

The ethical dilemmas are therefore likely to require countermeasures to preserve human autonomy and protect the integrity of our beliefs and values. The ideas selected by the author to support these values are briefly presented below.

### 4.1 Guidelines

In 2022, Brenner et al. [33] published six proposals for a more ethical design of persuasive systems based on a large systematic literature review and an interview study.

| Design Guideline | Description                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Awareness        | Creators should be aware of their ongoing asymmetric power dynamic to influence user behavior and decision making, which can have serious consequences for the user. |
| Outcomes         | The outcomes should be embedded within ethical boundaries that do not exploit users regardless of context, including for the public good.                            |
| Choice           | Give users sufficient autonomy when designing the choice architecture to ensure self-determination and freedom of choice.                                            |
| Autonomy         | Give the user freedom and autonomy to engage in the persuasion process through opt-in design.                                                                        |
| Transparency     | Balance transparency between the interests of users and the implications and effectiveness.                                                                          |
| Motivation       | Choose techniques that are fair, do not promote negative morality and present appropriate mechanisms in a visually appealing way.                                    |

Table 4: Propositions overview by [33].

Berdichevsky and Neuenschwander concisely and briefly formulated a golden rule of persuasion: "The creators of a persuasive technology should never seek to persuade a person or persons of something they themselves would not consent to be persuaded to do" [34].

## 4.2 Value Sensitive Design (VSD) Approach

Value Sensitive Design (VSD), developed by Friedman et al. [35], is a theoretical and methodological approach that aims to consider human values throughout the design process in a principled and comprehensive way. They define values in such a way that they direct attention to what is considered essential or of greatest importance to an individual or group. This also applies to interactive design. Technologies are not value-neutral, but rather reflect and shape the values and norms of society. VSD aims to encourage designers to think about the potential impact of their work on users and stakeholders from the outset and to incorporate ethical considerations into the development process. Friedmann et al. [35] have divided the main features of the approach into three parts.

- *Conceptual*: The first step is for the developer to identify the impact of the proposed technology on indirect and direct stakeholders. VSD divides attention among several stakeholder groups.
- *Empirical*: Studies should then be conducted to validate, develop and refine the concepts created.
- *Technical*: All the information obtained is used for the design.

## 4.3 Response to AI-driven persuasion

Burtell and Woodside [30] propose several potential responses to address the impact of AI-driven persuasion. However, they advise against banning persuasive AI, as persuasion works not only through verbal systems but also non-verbally. The diversity would therefore also lead to a restriction of a considerable part of AI. However, AI-driven systems could be limited to small groups of governments or companies that are considered responsible stewards of the technology, reducing risks and preserving benefits. Regulation of this type in practice, particularly on an international scale, is difficult. AI model providers, on the other hand, could develop a model-generated output or hide signatures in the output for automatic labeling of AI systems to give humans the ability to distinguish them. It is important to require and technically enforce the truthfulness or honesty of AI agents. Here, truthfulness determines an AI system's statements with reality and honesty determines its output with its own beliefs [36]. As AI systems become more powerful, it may become increasingly difficult for humans to verify the truthfulness of their statements. In these cases, reliance must be placed on truth-enhancing evolution. Sometimes, however, untruths can be an advantage, for example in the case of fiction or white lies. Honesty could certify a probability of truthfulness to the AI system through case law, models or humans in the development processes. Another way to mitigate problems of AI-driven persuasion is to utilize existing methods of liability and regulation. If humans have effective means to address problems caused by persuasive AI through the

liability system, this could greatly change the incentives of the creators and deployers of the systems. However, this would also require different regulatory statics, as it would be difficult for the persuader to distinguish between human-driven and ai-driven content [30].

#### **4.4 Test for ethically acceptance**

If a persuasive system is already in use, there are five elements [21] that can be used to decide whether the technology is persuasive and ethically acceptable or manipulative and unethical. First, is there a disclosure of intent and clarity about the end goal and methodology of the influence? This should include consideration of whether the user is given the opportunity to actively weigh options against their values, beliefs and desires. Is there any form of psychological, emotional or behavioral weakness that exploits the user's values and beliefs? It is also important to determine whether the algorithm uses the user as a resource or respects their autonomy. And finally, is the user in control of the decision-making ability?

### **5 Conclusion**

Today's attention economy plays with behavior-altering systems that direct their product, the attention of users, with a wide variety of technologies. It became clear that ethical concerns must be discussed in this process. Persuasive design always has a dual role: it must guide users effectively and efficiently through the endless flood of information and must uphold ethical principles in order to respect the autonomy and dignity of users. A central challenge is to develop design strategies that are both effective and ethically justifiable. AI is of particular concern as it has the potential to revolutionize the dynamics of human persuasion and at the same time creates new ethical dilemmas and responsibilities. Selected guidelines, policies and questions for future users and designers have been presented.

This work has approached the topic from a more critical perspective. Yetim [37] argues that these valuable critical insights are based on a narrow and predominantly pessimistic view of modernity. He recommends the elaboration of Habermas [38], a critical analysis of modernism, who proposes theoretical concepts for a considered analysis and an ethically grounded design to realize unfulfilled ethical potentials of persuasive technologies.

The design ethicist Tristan Harris also repeatedly emphasizes that the power to decide on developments does not lie in the hands of people, but with companies [39].

However, the openness and sharing of information on this topic is in our own hands.

# Bibliography

- [1] H. A. Simon, “Designing organizations for an information-rich world,” *Computers, communications, and the public interest*, vol. 72, p. 37, 1971.
- [2] D. G. Ramirez, “Journalism in the attention economy: The relation between digital platforms and news organizations1,” *Brazilian journalism research*, vol. 17, pp. 4–27, 2022.
- [3] J. Williams, *Stand out of our light: Freedom and resistance in the attention economy*. Cambridge University Press, 2018.
- [4] R. A. Lanham, *The economics of attention: Style and substance in the age of information*. University of Chicago Press, 2006.
- [5] N. Srnicek, *Plattform-Kapitalismus*. Hamburger Edition HIS, 2018.
- [6] Center for Humane Technology, *When attention went on sale*, <https://www.humanetech.com/podcast/16-when-attention-went-on-sale>, [Podcast]. Your Undivided Attention, Episode 16, 2020.
- [7] S. Giraldo-Luque, P. N. Aldana Afanador, and C. Fernández-Rovira, “The struggle for human attention: Between the abuse of social media and digital wellbeing,” in *Healthcare*, MDPI, vol. 8, 2020, p. 497.
- [8] H. Oinas-Kukkonen and M. Harjumaa, “Towards deeper understanding of persuasion in software and information systems,” in *First international conference on advances in computer-human interaction*, IEEE, 2008, pp. 200–205.
- [9] B. J. Fogg, “Persuasive technology: Using computers to change what we think and do,” *Ubiquity*, vol. 2002, no. December, p. 2, 2002.
- [10] M. Harjumaa and H. Oinas-Kukkonen, “Persuasion theories and it design,” in *Persuasive Technology: Second International Conference on Persuasive Technology*, Springer, 2007, pp. 311–314.
- [11] H. Oinas-Kukkonen and M. Harjumaa, “Persuasive systems design: Key issues, process model, and system features,” *Communications of the Association for Information Systems*, vol. 24, pp. 485–500, Mar. 2009.
- [12] B. Cushman-Roisin. “Persuasive technology.” Accessed on [29.02.2024]. (2019), [Online]. Available: <https://cushman.host.dartmouth.edu/courses/engs44/PersuasiveTechnology.pdf>.
- [13] W. Lidwell, K. Holden, and J. Butler, *Universal principles of design, revised and updated: 125 ways to enhance usability, influence perception, increase appeal, make better design decisions, and teach through design*. Rockport Pub, 2010.

- [14] M. Weinmann, C. Schneider, and J. v. Brocke, “Digital nudging,” *Business & Information Systems Engineering*, vol. 58, pp. 433–436, 2016.
- [15] R. H. Thaler, C. R. Sunstein, and J. P. Balz, “Choice architecture,” *The behavioral foundations of public policy*, vol. 25, pp. 428–439, 2013.
- [16] E. J. Johnson, S. B. Shu, B. G. Dellaert, *et al.*, “Beyond nudges: Tools of a choice architecture,” *Marketing letters*, vol. 23, pp. 487–504, 2012.
- [17] R. Thaler and C. Sunstein, *Nudge: Improving Decisions about Health, Wealth, and Happiness*. Yale University Press, 2008. [Online]. Available: <https://books.google.de/books?id=dSJQn8egXvUC>.
- [18] M. Dehnert and P. A. Mongeau, “Persuasion in the age of artificial intelligence (ai): Theories and complications of ai-based persuasion,” *Human Communication Research*, vol. 48, no. 3, pp. 386–403, 2022.
- [19] G. Huang and S. Wang, “Is artificial intelligence more persuasive than humans? a meta-analysis,” *Journal of Communication*, vol. 73, no. 6, pp. 552–562, 2023.
- [20] D. Susser, B. Roessler, and H. Nissenbaum, “Online manipulation: Hidden influences in a digital world,” *Geo. L. Tech. Rev.*, vol. 4, p. 1, 2019.
- [21] M. Botes, “Autonomy and the social dilemma of online manipulative behavior,” *AI and Ethics*, vol. 3, no. 1, pp. 315–323, 2023.
- [22] S. Zuboff, “The age of surveillance capitalism,” in *Social theory re-wired*, Routledge, 2023, pp. 203–213.
- [23] F. Lupiáñez-Villanueva, A. Boluda, F. Bogliacino, G. Liva, L. Lechardoy, and T. R. de las Heras Ballell, *Behavioural study on unfair commercial practices in the digital environment: dark patterns and manipulative personalisation*. Publications Office of the European Union, 2022.
- [24] “Report on the right to privacy in the digital age,” United Nations, Tech. Rep., 2018, p. 2.
- [25] A. Stibe and B. Cugelman, “Persuasive backfiring: When behavior change interventions trigger unintended negative outcomes,” in *Persuasive Technology: 11th International Conference, Proceedings 11*, Springer, 2016, pp. 65–77.
- [26] A. Harris, J. Qadir, U. A. Khan, *et al.*, “Persuasive technology for human development: Review and case study,” *arXiv preprint arXiv:1708.08758*, 2017.
- [27] N. Jacobs, “Two ethical concerns about the use of persuasive technology for vulnerable people,” *Bioethics*, vol. 34, no. 5, pp. 519–526, 2020.
- [28] T. Nyström and A. Stibe, “When persuasive technology gets dark?” In *European, mediterranean, and middle eastern conference on information systems*, Springer, 2020, pp. 331–345.

- [29] IBM, *What is artificial intelligence (ai)?* <https://www.ibm.com/topics/artificial-intelligence>, Accessed [24.03.2023], 2023.
- [30] M. Burtell and T. Woodside, “Artificial influence: An analysis of ai-driven persuasion,” *arXiv preprint arXiv:2303.08721*, 2023.
- [31] B. J. Dietvorst, J. P. Simmons, and C. Massey, “Algorithm aversion: People erroneously avoid algorithms after seeing them err.,” *Journal of Experimental Psychology: General*, vol. 144, no. 1, p. 114, 2015.
- [32] M. Guerini, F. Pianesi, and O. Stock, “Is it morally acceptable for a system to lie to persuade me?” In *Workshops at the twenty-ninth AAAI conference on artificial intelligence*, 2015.
- [33] D. Benner, S. M. Schöbel, A. Janson, and J. M. Leimeister, “How to achieve ethical persuasive design: A review and theoretical propositions for information systems,” *AIS Transactions on Human-Computer Interaction*, vol. 14, no. 4, pp. 548–577, 2022.
- [34] D. Berdichevsky and E. Neuenschwander, “Toward an ethics of persuasive technology,” *Communications of the ACM*, vol. 42, no. 5, pp. 51–58, 1999.
- [35] B. Friedman, P. Kahn, and A. Borning, “Value sensitive design: Theory and methods,” *University of Washington technical report*, vol. 2, no. 8, 2002.
- [36] O. Evans, O. Cotton-Barratt, L. Finnveden, *et al.*, “Truthful ai: Developing and governing ai that does not lie,” *arXiv preprint arXiv:2110.06674*, 2021.
- [37] F. Yetim, “Critical perspective on persuasive technology reconsidered,” in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2013, pp. 3327–3330.
- [38] J. Habermas, *The theory of communicative action: Volume 1: Reason and the rationalization of society*. Beacon press, 1985, vol. 1.
- [39] T. Harris, “How a handful of tech companies control billions of minds every day,” *Los Angeles, California, Estados Unidos. Obtenido de <https://www.youtube.com/watch>*, vol. 74, 2017.